

Ali Elaswad

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EDUCATION

The American University in Cairo (AUC)

Sep 2022 – Dec 2026

B.Sc. in Computer Engineering

GPA: 3.96/4.0

Concentration: Artificial Intelligence · **Minor:** Business Administration

Relevant Courses: Deep Learning, Machine Learning, Data Science Practice and Research, Probability and Statistics, Distributed Systems, Applied Quantum Computing

PUBLICATIONS

[1] **Decision-Focused Learning for Operating Room Scheduling Under Uncertain Surgery Durations with Two-Stage Stochastic Optimization.**

Ali Elaswad, Rodrigo Carrasco, Nourhan Sakr. *ECML PKDD 2026*, oral.

[2] **Articulator-Level Cross-Lingual Transfer in Sign Language Translation.**

Ali Elaswad, Omar Saqr. *WSLP @ EMNLP 2026*, oral.

[3] **An Ensemble of Encoders With Agentic Error Refinement.**

Noor Emam, Omar Saqr, Mostafa Gaafar, Ali Elaswad. *ArabicNLP 2026*, 1st Place in 6 of 8 tracks.

[4] **GeoDrive: Cross-City Autonomous Driving through Meta-Learning and LLMs.**

Mostafa Gaafar*, Ali Elaswad*, Hamada Rizk, Moustafa Youssef. *ACM SIGSPATIAL 2025 SRC*, 1st Place, oral.

[5] **Stage-Fair Electric Vehicle Charging Infrastructure Planning under Coverage-Adaptive Demand.**

Ali Elaswad, Nourhan Sakr. *Under review*.

[6] **Data-Efficient Cross-City Adaptation for Autonomous Driving.**

Ali Elaswad*, Mostafa Gaafar*, Hamada Rizk, Moustafa Youssef. *Under review*.

*Equal contribution.

EXPERIENCE

Microsoft

Jul 2026 – Aug 2026

Applied Scientist Intern, Microsoft AI (Shopping)

- Developed a Qwen3-0.6B aspect-based sentiment model, distilled from **700k** GPT-5.5-labeled product reviews
- Raised F1 from **53% to 75%** and extracted **88%** more opinions per review at **95%** precision
- Replaced a two-stage BERT pipeline with one generative pass that still meets production latency and QPS
- Optimized an LLM judge for Microsoft-to-Shopify taxonomy mapping with GEPA, gaining **10%** accuracy
- Migrated the **25.7B-row** (2.8 TB) review pipeline to Spark with exact parity on all **97** outputs

University of Waterloo

Feb 2026 – Present

Research Intern

Advisors: Prof. Raouf Boutaba, Prof. Noura Limam

- Developed a PPO agent that acts as the 5G Policy Control Function, closing the QoS loop from core telemetry
- Designed a multi-discrete action space over GBR/MBR limits and flow preemption for **3** traffic classes
- Engineered a min-min reward over per-flow throughput, loss, and delay margins so no single flow gets starved
- Built a labeled degradation dataset and benchmarked the agent across **7** congestion scenarios
- Building a scheduling gym (token buckets, queueing) with a data-center case study on real packet traces
- Preparing a paper for IEEE NOMS; the project is also my AUC senior thesis

Osaka University

Jun 2025 – Jun 2026

Deep Learning Research Intern

Advisors: Prof. Moustafa Youssef, Prof. Hamada Rizk

- Co-developed GeoDrive, pairing MAML with OpenStreetMap context so driving policies adapt to new cities fast
- Improved trajectory prediction by **18%** within a city and **32%** across cities on nuScenes

- Distilled Gemini 2.5 Pro into an on-device SmolLM with LoRA for interpretable, region-aware decisions
- Won **1st place** at the ACM SIGSPATIAL 2025 SRC

The American University in Cairo

2025 – Present

Undergraduate Researcher

Advisors: Prof. Nourhan Sakr; Prof. Rodrigo Carrasco

- Developed an SPO+ and two-stage stochastic recourse framework for OR scheduling on real neurosurgery data
- Reached **94.4%** OR utilization and cut 95th-percentile overtime by **36%** (2.21h to 1.41h)
- Formulated a fairness-constrained, rolling-horizon MIP for staged EV charger rollout across Georgia
- Delivered **2.6×** more zones within 2 km of a charger and **31%** shorter access distance vs. DOVA-PATBM

HONORS AND AWARDS

- **1st Place, ACM SIGSPATIAL 2025 Student Research Competition**, Undergraduate Category 2025
- **AUC Award for Excellence in Undergraduate Research** 2026
- **AUC School of Sciences and Engineering Honors Assembly Awardee**, top 1% of senior students 2026
- **2x AUC Undergraduate Research Grant**, funding for the research projects
- **AUC Dean’s List**, 7 consecutive semesters 2023 – 2026

TEACHING

The American University in Cairo

Fall 2024 – Spring 2025

Teaching Assistant, Dept. of Computer Science and Engineering

- Guided students through Verilog and FPGA design in Computer Architecture Lab
- Supported Digital Design Lab and Discrete Mathematics sections (Fall 2024)

SKILLS

Programming: Python, C++ , Java, Rust, JavaScript, SQL, Verilog
ML & NLP: PyTorch, TensorFlow, Hugging Face, LoRA fine-tuning, LLM distillation, prompt optimization
Data & Systems: Apache Spark, Pandas, Docker, Kubernetes, MPI, Linux, Git, ~~TeX~~
Athletics: Member, AUC Tennis Team.